

# AWS EC2 vs Lambda Practical Beginner Cheat Sheet

Choosing between EC2 and Lambda determines how you build, run, and pay for your cloud applications. Understanding when to rent a virtual server versus running code on demand will save you time and prevent unexpected cloud bills.

### What you'll learn:

- 1 Launching an EC2 Virtual Machine
- 2 Creating a Serverless Lambda Function
- 3 Testing Lambda Functions via CLI
- 4 Streaming Real-Time Lambda Logs

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# 1

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## Launching an EC2 Virtual Machine

Amazon EC2 gives you full control over a virtual computer in the cloud. You select the operating system, compute power, and network settings, making it perfect for persistent web servers or long-running applications that must stay active continuously.

```
$aws ec2 run-instances --image-id ami-0c55b159cbfafe1f0 --count 1 --instance-type t2.micro --key-name
```

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# 2

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## Creating a Serverless Lambda Function

AWS Lambda lets you execute code without provisioning or managing underlying servers. You upload your code, define execution permissions, and AWS handles automatic scaling and server upkeep behind the scenes.

```
$aws lambda create-function --function-name ProcessOrder --runtime python3.9 --role arn:aws:iam::1234
```

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# 3

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## Testing Lambda Functions via CLI

Before deploying your serverless functions to production, you can trigger them locally using the AWS Command Line Interface. Invoking the function sends test payloads directly and saves the execution output to a local JSON file for inspection.

```
$aws lambda invoke --function-name ProcessOrder --payload '{"order_id": 101}' response.json
```

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# 4

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## Streaming Real-Time Lambda Logs

Because serverless environments lack terminal access to the host machine, debugging relies on central logging. CloudWatch automatically captures print statements and errors, which you can stream directly to your terminal screen in real time.

```
$aws logs tail /aws/lambda/ProcessOrder --follow
```



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## Key Takeaways

- + Use Lambda for unpredictable or event-driven tasks to avoid paying for idle computing time.
- + Choose EC2 when your workload runs continuously or takes longer than Lambda's 15-minute limit.
- + Assign minimum required IAM permissions to both EC2 instances and Lambda functions.
- + Set up CloudWatch billing alarms to monitor unexpected resource usage early.
- + Use Lambda environment variables to manage configuration values securely.

## Watch Out For

- ! Leaving idle EC2 instances running continuously and incurring unintended hourly charges.
- ! Forgetting that Lambda functions automatically force-stop after a maximum timeout of 15 minutes.
- ! Expecting files saved in Lambda's local temporary storage (/tmp) to persist permanently across executions.
- ! Selecting oversized EC2 instance types for small projects without analyzing actual CPU usage.

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